ANALYSIS REPORT

TE Curve Verification Pipeline Directional Model Comparison

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report is the canonical `TE Curve Verification Pipeline` offline comparison between `RCIM Model-Bank Reproduction`, recovered original, retuned paper-reference model banks, and repository-owned `Wave 1` and `Wave 2.1` model candidates. It starts from the current direction-aware comparison matrix.

Dataset And Split

- dataset config: `config/datasets/transmission_error_dataset.yaml`;
- dataset root: `data\simplified_dataset`;
- comparison mode: `reduced_selected_model_matrix`;
- candidate count: `4`;
- held-out curve count before candidate filtering: `97`;
- percentage-error denominator: `peak_to_peak_truth`;
- `Fw` candidates are evaluated only on forward curves;
- `Bw` candidates are evaluated only on backward curves;

Candidate Inventory

Candidate	Source	Family	Surface	Valid Directions
`simplified_feedforward_Bw`	`simplified_legacy_model_development_registry`	`feedforward`	`Bw`	`backward`
`simplified_tree_Bw`	`simplified_legacy_model_development_registry`	`tree`	`Bw`	`backward`
`simplified_harmonic_regression_Bw`	`simplified_legacy_model_development_registry`	`harmonic_regression`	`Bw`	`backward`
`simplified_periodic_gru_sequence_Bw`	`simplified_legacy_model_development_registry`	`periodic_gru_sequence`	`Bw`	`backward`

Curve Evidence

Each selected candidate is shown with the same four deterministic held-out operating conditions for this direction. The dark line is the measured TE curve and the blue line is the model prediction for the same operating condition.

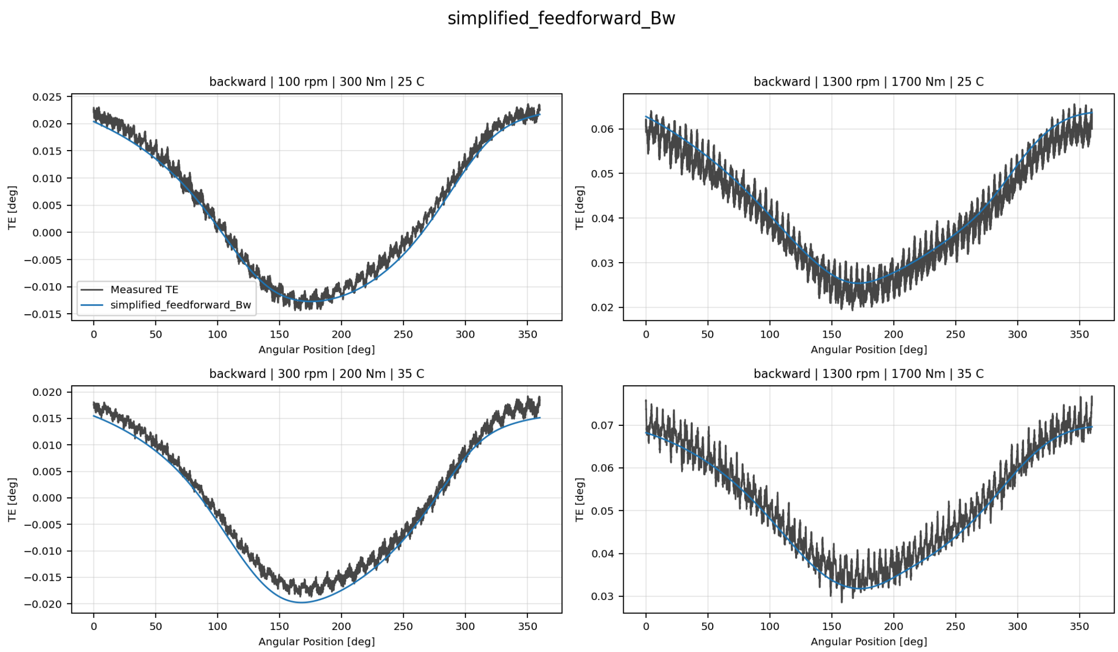
Shared operating conditions

- 100 rpm, 300 Nm, 25 C
- 1300 rpm, 1700 Nm, 25 C
- 300 rpm, 200 Nm, 35 C

- 1300 rpm, 1700 Nm, 35 C

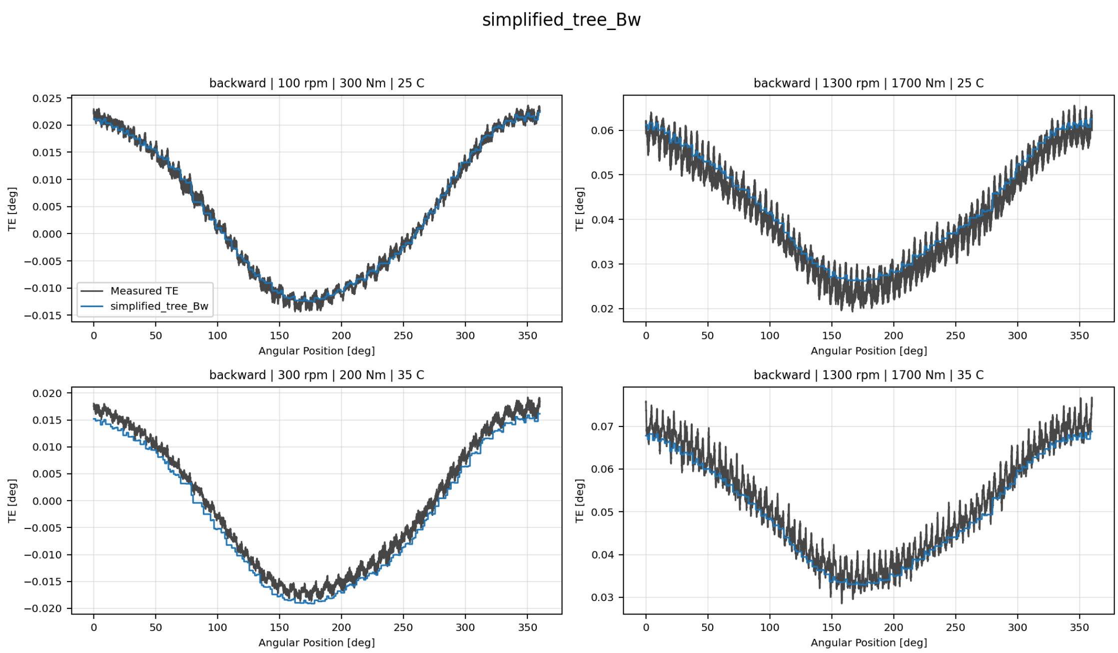
Candidate	Role	Curve MAE [deg]	Curve RMSE [deg]	Mean MPE [%]	P95 MPE [%]
`simplified_feedforward_Bw`	Selected candidate	0.003586	0.004023	7.832	14.856
`simplified_tree_Bw`	Selected candidate	0.003258	0.003651	7.051	14.116
`simplified_harmonic_regression_Bw`	Selected candidate	0.003678	0.004012	8.058	15.071
`simplified_periodic_gru_sequence_Bw`	Surface leader	0.002392	0.002639	5.466	14.820

simplified_feedforward_Bw



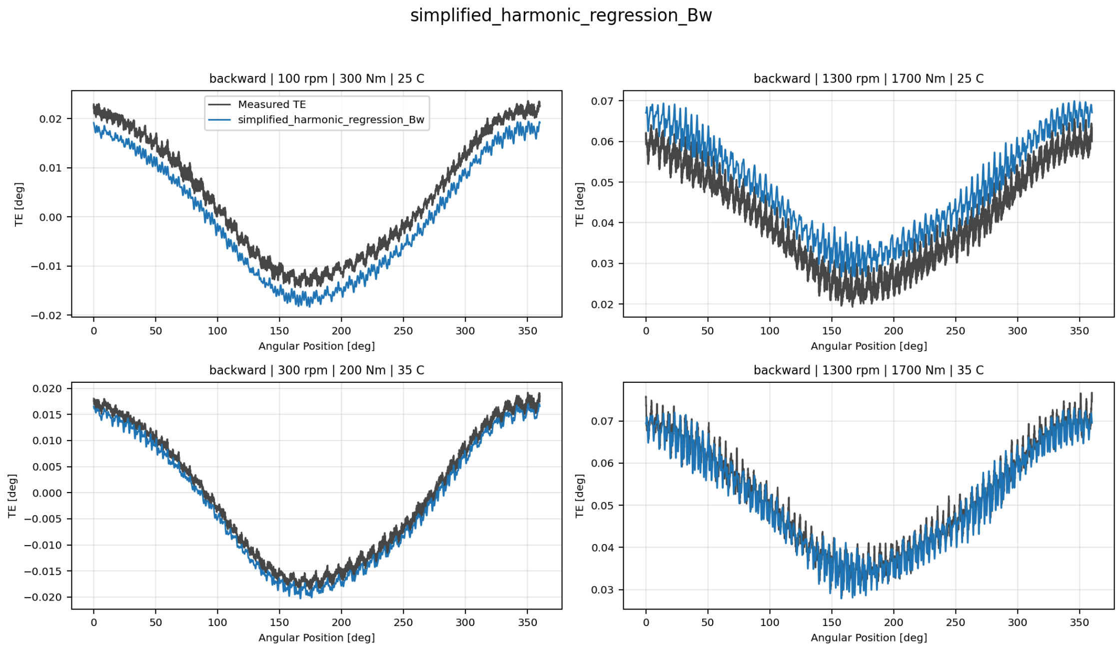
simplified_feedforward_Bw measured-versus-predicted TE collage

simplified_tree_Bw



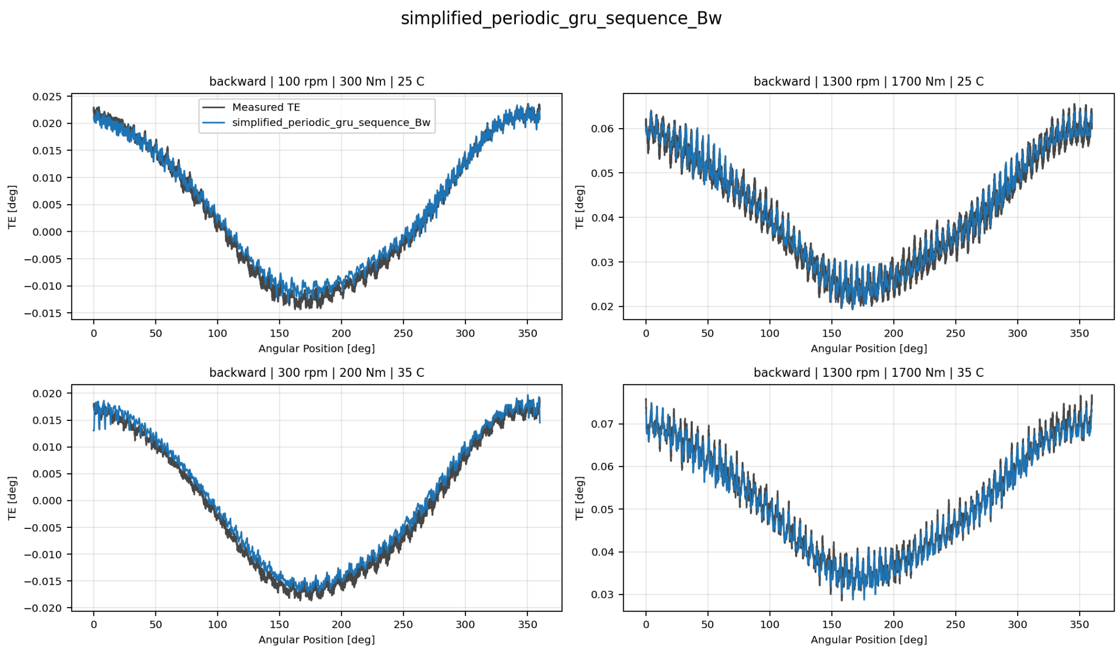
simplified_tree_Bw measured-versus-predicted TE collage

simplified_harmonic_regression_Bw



simplified_harmonic_regression_Bw measured-versus-predicted TE collage

simplified_periodic_gru_sequence_Bw



simplified_periodic_gru_sequence_Bw measured-versus-predicted TE collage

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-07-01-22-16__track2_reduced_selected_model_matrix_track2_selected_simplified_dataset_backward_2026_07_06/validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-07-01-22-16__track2_reduced_selected_model_matrix_track2_selected_simplified_dataset_backward_2026_07_06\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots\paused_reduced_selected_model_matrix` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within each source group and direction. Directional paper-reference, Wave 1, and Wave 2.1 models are never evaluated on the opposite direction. Global Wave models remain valid on both directions and are therefore shown in the directional sections and again in the global breakdown. The `rcim_track1` forward reference banks use the opposite stored `h0` sign convention relative to the TE Curve Verification Pipeline reconstruction contract, so the TE Curve Verification Pipeline comparison applies the documented source-specific `h0` compatibility multiplier before curve reconstruction.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online `Table 9` compensation benchmark.
- The report uses the saved Python model artifacts from `models/` ; ONNX parity checks remain a separate deployment-readiness task.